

CSE 3252: Web Engineering Lab

Credits	1
Contact Hours	28
Year	Third
Semester	Second
Prerequisite	CSE 2232: User Interface Development Lab
Course Type	☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce

Motivation

Web Engineering Lab complements the theory course by giving students hands-on experience in building full-stack web applications. Using modern frontend and backend frameworks, students will learn how to implement secure, database-driven, and maintainable web systems.

Course Objective

This laboratory course develops practical skills in full-stack web development using React on the frontend and a modern server-side web framework on the backend. Students will design APIs, connect databases, implement authentication, integrate frontend and backend modules, test web applications, and package them using modern development tools.

Course Outcomes (COs), Program Outcomes (POs) and Assessment

CO No.	CO Statement	Corresponding PO	Domain / Level of Learning Taxonomy	Delivery Methods & Activities	Assessment Tools
CO1	To build RESTful backend services and database-driven features using a modern server-side web framework	Modern Tool Usage (PO5)	Cognitive domain – Level 6	☐ Lecture ☒ Note ☒ Text Book ☐ ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☐ Final Exam ☒ ☐ Assignment ☒ Notebook ☒ Presentation
CO2	To integrate React frontends with backend APIs, authentication, validation, and persistence to develop full-stack web applications	Design/Development of Solutions (PO3)	Cognitive domain – Level 6	☐ Lecture ☒ Note ☒ Text Book ☐ ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☐ Final Exam ☒ ☐ Assignment ☒ Notebook ☒ Presentation
CO3	To test and package secure web applications using Postman, Git, Docker, and modern framework toolchains	Modern Tool Usage (PO5)	Cognitive domain – Level 6	☐ Lecture ☒ Note ☒ Text Book ☐ ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ ☐ Assignment ☒ Notebook ☒ Presentation

Assessment and Marks Distribution

Students will be assessed on the basis of their overall performance across all lab exercises, assignments, and the final lab examination.

Component	Weightage
Continuous Assessments (CA) — lab exercises and assignments throughout the semester	20%
Comprehensive Final Exam + Lab Notebook	70%
Class Participation	10%

Lab Course Contents / List of Experiments

Lab 1 — Environment Setup and Full-Stack Project Structure

- Install and configure Node.js, Composer, PHP, MySQL/PostgreSQL, Postman, and Docker Desktop
- Create a React frontend with Vite and initialize a backend project in the chosen server-side framework
- Understand the full-stack folder structure, client-server interaction, and API-first workflow

Lab 2 — Server-Side Framework Fundamentals

- Project structure, routing, controllers, middleware, and configuration in the selected backend framework
- Database migrations, models, seeders, and ORM / query builder basics
- Build a simple resource module with database-backed CRUD

Lab 3 — REST API Development

- Build RESTful API endpoints with controllers and resource routes
- Request validation, JSON responses, error handling, and API testing with Postman
- Model relationships and relational CRUD operations

Lab 4 — Alternative Backend Framework / Comparative Implementation

- Implement the same or a similar backend module in an alternative server-side framework if taught during the semester
- Compare routing, middleware, validation, ORM / query builder usage, and project structure

- Explore similarities and differences between backend framework workflows

Lab 5 — Authentication and Authorization

- User registration, login, logout, password hashing, protected routes
- Token/session-based authentication concepts in the chosen backend framework
- Role-based access and route protection for admin/user features

Lab 6 — React Frontend Integration

- Connect React components to backend APIs using Fetch or Axios
- Form submission, loading states, validation messages, and error handling
- React Router integration with public and protected pages

Lab 7 — Full-Stack CRUD Module

- Build a complete CRUD module with React frontend and backend API
- Search, filter, pagination, and relational data display
- Keep frontend state synchronized with server updates

Lab 8 — File Uploads, Security, and Validation

- File/image upload workflow and storage handling
- Input validation and sanitisation at frontend and backend levels
- Basic lab-level security practices: authentication checks, protected resources, and secure error handling

Lab 9 — Testing, Debugging, and API Documentation

- Create Postman collections for endpoint testing
- Basic automated API tests / request validation checks
- Debugging with browser DevTools, backend logs, and framework debugging tools
- Prepare API usage documentation for the final project

Lab 10 — Docker and Capstone Full-Stack Project

- Dockerise the application stack using containers for frontend, backend, and database
- Use Docker Compose for local development and service orchestration

- Final project: develop and demonstrate a secure full-stack web application using React frontend and the selected backend framework

Capstone requirements: - User authentication - Database-backed CRUD operations - React frontend integrated with backend API - Validation and error handling - Postman collection / API documentation - Docker-based local setup

Textbooks

1. Matt Stauffer — *Laravel: Up & Running* (3rd ed.), O'Reilly Media, 2023
2. Ethan Brown — *Web Development with Node and Express* (2nd ed.), O'Reilly Media, 2019

Recommended Resources

1. **Laravel Documentation** — laravel.com/docs
2. **Express.js Documentation** — expressjs.com
3. **AdonisJS Documentation** — docs.adonisjs.com
4. **React Documentation** — react.dev
5. **Postman Learning Center** — learning.postman.com
6. **Docker Documentation** — docs.docker.com

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